

Annual Space Environment Report: Orbital Debris and LEO Congestion

Orbital Sustainability Office (ficticio, estilo ESA Space Debris Office)

Overview of the current environment

Space surveillance networks now track tens of thousands of objects larger than ten centimeters in Earth orbit, a small fraction of the much larger population of debris fragments too small to be individually cataloged but still capable of damaging operational spacecraft.

The rate of launches into low Earth orbit has grown sharply in recent years, driven largely by the deployment of large satellite constellations for broadband connectivity, which has accelerated concerns about long-term congestion in the most heavily used orbital shells.

Sources of debris

Fragmentation events, whether from collisions or from the break-up of spent rocket stages, remain the dominant source of trackable debris. Each fragmentation event can generate thousands of new pieces, many of which persist in orbit for decades.

Intentional destructive testing of satellites has historically produced some of the largest single increases in tracked debris population, prompting renewed international calls to restrict such tests.

Mitigation measures

Post-mission disposal guidelines recommend that satellites in low Earth orbit re-enter the atmosphere within a bounded number of years after the end of operations, though compliance across operators remains uneven.

Active debris removal demonstrations, still at an early technological readiness level, are being evaluated as a complementary measure for objects that cannot deorbit passively within an acceptable timeframe.

Outlook for orbital sustainability

Projections included in this report suggest that without stronger adherence to mitigation guidelines, the density of debris in the most congested shells could reach a threshold where collisions generate debris faster than natural atmospheric drag removes it, a scenario long referred to in the technical literature as a cascading instability.

International coordination among space agencies and commercial operators is identified as the single most important factor in avoiding that outcome over the coming decade.